

Antidepressant effect of Chaihu-Shugan-San extract and its constituents in rat models of depression

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Abstract

Herbal preparations may be effective alternatives in the treatment of depression, which remains difficult to manage. Chaihu-Shugan-San (CSS), an oriental traditional medicine, has been used as a remedy for Hwa-Byung, a Korean culture-bound syndrome resembling depression. We examined whether aqueous extracts of CSS and its constituent herbs exert antidepressant-like effects in two experimental animal models: the forced swimming test (FST) and the chronic mild stress (CMS) model. The herbal extracts were administered orally for 7 days in the FST and for 21 days during the CMS model; imipramine at 20 mg/kg/day was injected intraperitoneally as a positive control. CSS, Radix Bupleuri (one of the most important constituent plants in CSS), and imipramine had significant anti-immobility effects in the FST and reversed the CMS-induced reduction in sucrose consumption. Rhizoma Cyperi, another constituent of CSS, had antidepressant activity in the FST, while it failed in the CMS model. In conclusion, our results suggest that CSS and its constituent herbs exert antidepressant-like effects comparable to those of imipramine in experimental animal models.

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Introduction

Depression is a common illness associated with high rates of chronicity, relapse, and recurrence; psychosocial and physical impairment; and a high suicide rate. Nonetheless, many currently available antidepressants have low rates of response and remission. Moreover, contemporary antidepressants can produce many unwanted side effects. In Oriental society, herbal preparations are widely used by consumers and have a long history of use as medicines. Some of them may be effective alternatives in the treatment of depression, as in the case of St John's wort, a Western herb (Müller, 2003).

In order to screen traditional herbal formulas for antidepressant effects, it is necessary to find a folk illness that is similar to depression. Hwa-Byung (HB) has been categorized as a Korean culture-bound syndrome that refers to insidious, long-standing, serious discontent that is projected into the body and is manifested by numerous symptoms, such as insomnia, fatigue, panic, palpitations, and dyspnea (Park et al., 2001). It has been suggested that HB is a culturally patterned way of expression for Koreans experiencing major depression and related conditions (Lin et al., 1992). According to the concept of Korean folk illness, repression of anger (Hwa) and distress results in liver-qi stagnation (*gan-qi-yu-jue*), and this is thought to cause HB. Chaihu-Shugan-San (CSS), an oriental herbal medicine, is popularly used in folk medicine for relieving a wide variety of symptoms caused by liver-qi stagnation. Therefore, we postulated that CSS might have antidepressant activity and sought evidence of this experimentally. This study evaluated the antidepressant effects of CSS and its constituent herbs in two animal models of depression: the forced swimming test (FST) (Porsolt et al., 1978) and the chronic mild stress (CMS) model (Muscat et al., 1990).

Materials and methods

Preparation of extracts

The medicinal plants used to prepare CSS are listed in Table 1 (Ban et al., 1991). They were purchased from Myungji Oriental Material Medical Company (Gyeongju, Korea) and authenticated by Professor S.D. Park, College of Oriental Medicine, Dongguk University, based

Table 1
Composition of Chaihu–Shugan–San (CSS)

Medicinal Plant	Amount (g)
Radix Bupleuri	8.0
Aurantii nobilis Pericarpium	8.0
Rhizoma Cnidii	6.0
Rhizoma Cyperi	6.0
Radix Paeoniae	6.0
Poncirus trifoliata Rafinesque	6.0
Radix Glycyrrhizae	2.0

on their microscopic and macroscopic characteristics. The quality of these crude drugs is controlled by the Korean Pharmacopoeia 7th edition (1997), and a voucher specimen of each plant was prepared and deposited in the herbarium of the Department of Prescriptionology, Dongguk University.

The plant materials were cut into small pieces and then sun-dried. Dried samples weighing about 210 g were boiled in 1.5 L of distilled water for 3 h. After extraction, the decoction was collected, and the residue was subjected to further extraction in 1 L of boiling water for 3 h. The decoctions collected from the two successive extractions were mixed and passed through filter paper (Advantec Toyo, Japan). The filtrate was collected and concentrated in a water bath kept below 60°C under reduced pressure using a rotary evaporator (model RE440-WV, Yamato Scientific Co., Japan). The yields of CSS, Radix Bupleuri (BU), Aurantii nobilis Pericarpium (AU), Rhizoma Cnidii (CN), Rhizoma Cyperi (CY), Radix Paeoniae (PA), Poncirus trifoliata Rafinesque (PO), and Radix Glycyrrhizae (GL) were 28.9, 19, 23.6, 29.3, 17, 20.5, 33.1, and 28%, respectively. The preparation of the aqueous extract described here is essentially the same as that of traditional CSS.

Animals

Male Sprague-Dawley (SD) rats (160–180 g) were used in the FST. All the animals used in this study were cared for and treated humanely according to the Principles of Laboratory Animal Care (NIH Publication No. 85-23, revised 1985) and the Institutional Review Board of Inje University. The animals were housed in groups of six under controlled conditions of light (12-h light/dark cycle, lights on at 7:00 a.m.) and temperature ($25 \pm 1^\circ\text{C}$). Food and water were available ad libitum. For the FST, the animals were randomly assigned to one of ten equal groups: a group for CSS, seven groups for constituent herbs of CSS, a control group, and a group for imipramine. In the CMS model, 28 SD rats (240–260 g) were housed singly in cages and maintained in the same environment as that used for the FST. For the CMS model, five experimental groups were included: a group for CSS, two groups for active constituents of CSS in the preceding study (the FST), a control group, and a group for imipramine. Each experimental group consisted of ten rats. All the animals were purchased from Daehan Biolink Co. Ltd. (Eumsung, Korea).

Forced Swimming Test (FST)

The FST procedure has been described in detail elsewhere (Porsolt et al., 1978). Briefly, the rats were subjected to two trials during which they were forced to swim in a glass aquarium (diameter: 18 cm, height: 40 cm) containing 15 cm of water at 25°C. There was a 7-day interval between the first and second trials. The first trial lasted 15 min; the second lasted 5 min. “Swimming” was defined by escape behaviors, i.e., diving, rigorous paddling with all four legs, circling the tank, and clambering at the walls. “Immobility” was scored as floating and treading water just enough to keep the nose above water. The water was changed after each rat. Following the test, the animals were dried in a heated enclosure. During the test session, the immobility time was recorded using a video camera. Two observers who were blind to the kind of treatment each animal had received evaluated the tapes.

Measuring locomotor activity

In order to detect any association of immobility in the FST with changes in motor activity, the activities of animals treated with the extracts were tested in an open field. Each rat was placed in the center of the open-field apparatus, and the locomotor activity was assessed immediately before the FST. The open-field apparatus was a field, 70 cm in diameter, which was demarcated into 18 approximately equal areas. Hand-operated counters were used to score locomotion (number of line crossings within 5 min) and rearing frequencies (number of times an animal stood on its hind legs). A separate researcher, who was blind to the treatment group, scored the behavior in the open field. In this experiment, the animals received the same drugs and doses as those used when measuring immobility. The open-field apparatus was washed with a deodorant solution (0.5% ammonia, 15% ethanol, 10% extran, 5% isopropanol, 10% pinol, and 59.5% water) before each behavioral test to eliminate possible odor clues left by previous subjects. Experiments were performed in a dark room, and the apparatus was illuminated by a 60 W bulb positioned 1 m above the center of the circle.

Chronic mild stress (CMS) model and sucrose consumption test

Initially, the animals were trained to consume a palatable weak (1%) sucrose solution. They were given 1% sucrose solution for a 48-h period in their home cages with no food or water following food and water deprivation for 20 h. Subsequently, they were given sucrose for 1 h per day on five consecutive days. The sucrose intake was measured at the end of the training in order to group the rats. The rats in the experimental groups were then subjected to CMS for 6 weeks. Each week, the stress regimen followed that of Muscat et al. (1990) and consisted of the following stressors: 1) one 17-h period in a soiled cage, 2) three periods (7, 9, and 17 h) of low intensity stroboscopic illumination, 3) two periods of continuous overnight illumination, 4) two periods (7 and 17 h) of 45° cage tilt, 5) two 20-hr periods of food and water deprivation, 6) one 2-h period of restricted access to food, 7) two periods (3 and 5 h) of intermittent white noise (85 dB), 8) one 16-h period of water deprivation, and 9) one 1-h exposure to an empty bottle. A 1-h sucrose consumption test (SCT) was carried out following a 20-h period of food and water deprivation once each week. The sucrose consumption was recorded by reweighing pre-weighed bottles of test solution. The normal control group was housed in a separate room and was not subjected to any stress. The animals were deprived of food and water before the sucrose intake test. Otherwise, food and water were freely available in their home cages.

Experimental groups and drug administration

The drugs were dissolved in distilled water. Herbal extracts were administered orally at a volume of 10 ml/kg. The daily doses of CSS, BU, AU, CN, CY, PA, PO, and GL were 2,000, 510, 630, 590, 340, 410, 660, and 190 mg/kg, respectively. As a positive control, 20 mg/kg/day of imipramine (IMI; Sigma) was injected intraperitoneally in a volume of 1 ml/kg for the FST. Control animals received vehicle (VEH; distilled water) only. For the FST, the drugs were given 30 min after the first trial and then daily for seven consecutive days; the rats were subjected to the second trial 24 h after the last injection on Day 7. For the CMS model, all the drugs were given

Table 2

Effects of 7-day treatments with CSS, its ingredients, and imipramine on the open field parameters recorded for 5 min in rats

Treatment	Crossing	Rearing
VEH	53.7 $\pm$ 4.7	7.7 $\pm$ 2.4
IMI	57.6 $\pm$ 6.9	6.9 $\pm$ 1.3
CSS	59.1 $\pm$ 5.2	6.3 $\pm$ 1.9
BU	56.1 $\pm$ 7.4	7.0 $\pm$ 1.5
AU	47.1 $\pm$ 8.0	8.2 $\pm$ 2.9
CN	49.8 $\pm$ 5.3	6.7 $\pm$ 1.7
CY	60.1 $\pm$ 7.9	7.9 $\pm$ 1.8
PA	50.9 $\pm$ 6.1	7.5 $\pm$ 1.6
PO	49.4 $\pm$ 6.1	6.1 $\pm$ 1.7
GL	50.5 $\pm$ 5.9	7.9 $\pm$ 2.0

Each value represents the mean $\pm$ SEM for ten animals ($p > 0.05$; Kruskal–Wallis Test).

daily at 17:00 h since the fourth week; weekly sucrose consumption tests were carried out 24 h after the last drug administration.

Statistical analysis

The raw data were analyzed statistically. All the results are expressed as the means $\pm$ SEM. Non-parametric tests were used for all statistical comparisons. Overall differences according to the treatment were confirmed using the Kruskal–Wallis test, and then pairs of experimental groups were compared using the Mann–Whitney U test. Statistical differences were considered

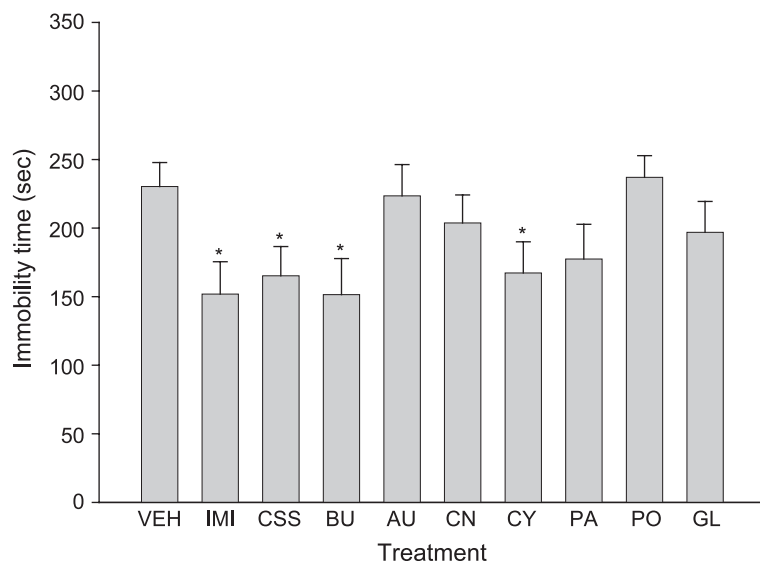


Fig. 1. Effects of CSS and its constituents on the FST in rats. The immobilization time (sec) was measured for the first 5 min after each animal was placed in the water. Each column represents the mean $\pm$ SEM ($n = 10$). The Mann–Whitney U test was performed on the raw data, $*p < 0.05$ vs. vehicle-treated group (VEH).

significant at $p < 0.05$. All analyses were conducted using the SPSS program for IBM-compatible computers.

Results

Effects of CSS and its constituents in the FST

Table 2 summarizes the open field behavior results. Locomotion (the number of crossings) ($\chi^2 = 4.28$, $p = 0.89$) and rearing frequencies ($\chi^2 = 1.28$, $p = 1.00$) did not differ statistically among the VEH-, CSS-, and constituent-treated rats. The behavioral effects produced in the FST owing to drug administration are presented in Fig. 1. The Kruskal-Wallis test showed a significant effect of drug treatment ($\chi^2 = 18.544$, $df=9$, $p = 0.029$). A significant reduction in immobility time was observed in the rats treated with IMI (151.9 ± 74.58 s, $U = 18.0$, $p = 0.016$), CSS (165.2 ± 67.61 s, $U = 24.0$, $p = 0.049$), BU (151.5 ± 83.06 s, $U = 23.0$, $p = 0.041$), and CY (167.3 ± 71.89 s, $U = 23.0$, $p = 0.041$) versus VEH (230.3 ± 55.21 s) in the FST. The other constituents were ineffective in reducing the immobility time in the FST.

Effects of CSS, BU, and CY on sucrose consumption

CSS and two of its constituent herbs that were effective in the FST (BU and CY) were further evaluated using the CMS model. In the sucrose solution-training phase (baseline phase), sucrose consumption did not differ significantly among the groups ($\chi^2 = 5.595$, $df = 8$, $p = 0.231$) (data not shown). The mean sucrose intake in the last baseline test was 12.3 ± 0.3 g. In the stressed rats, the reduction of sucrose consumption was significant after the first week of stress exposure, but reached

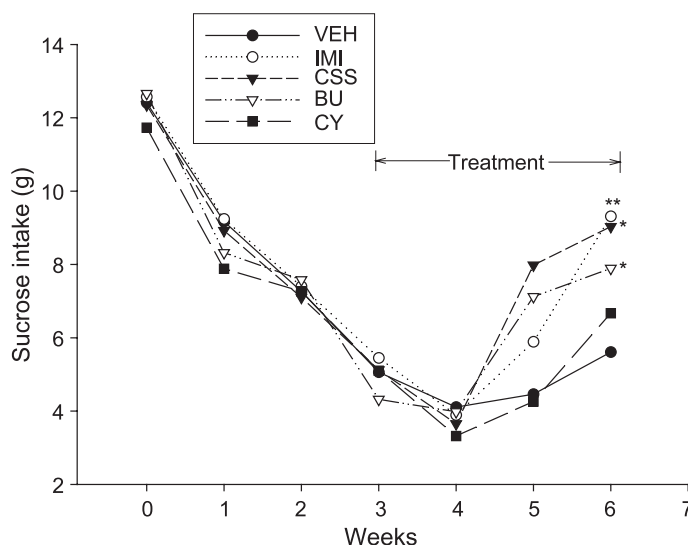


Fig. 2. Sucrose consumption in stressed animals treated daily with CSS, BU, and CY. The values are means ($n = 8$). Statistically significant differences indicated by $*p < 0.05$ and $**p < 0.01$ for drug-treated stressed groups are relative to vehicle-treated stressed animals (VEH) at the indicated time points.

a maximum during the fourth week ($U = 0.000$, $p = 0.001$) (Fig. 2). In the stressed animals, treatment with the experimental drugs caused a gradual recovery in sucrose intake compared with the sucrose intake at week 4. Stressed animals treated with IMI ($U = 1.0$, $p = 0.003$), CSS ($U = 6.0$, $p = 0.034$), and BU ($U = 14.5$, $p = 0.048$) for 3 weeks showed significant recovery compared with the vehicle-treated stressed animals (VEH) (Fig. 2).

Discussion

To our knowledge, these results are the first to demonstrate that CSS and some of its constituent herbs produce significant antidepressant-like effects comparable to those of imipramine in two animal models: the FST and the CMS model. Therefore, CSS or its constituents might have antidepressant actions in humans. Both CSS and BU, one of the most important constituents of CSS, caused a significant anti-immobility effect in the FST and reversed the CMS-induced reduction in sucrose consumption. CY, another constituent of CSS, had antidepressant activity in the FST, while it failed in the CMS model.

The FST is the tool most widely used for assessing antidepressant activity preclinically. The widespread use of this model is largely a result of its ease of use, reliability across laboratories, and ability to detect a broad spectrum of antidepressant agents (Borsini and Meli, 1988). Most clinically active antidepressants are effective in the FST, while neuroleptics and anxiolytics produce different effects (Porsolt et al., 1979). Antidepressants can also be distinguished from stimulants because stimulants cause marked motor stimulation, in contrast to antidepressants (Borsini and Meli, 1988). In our study, the antidepressant-like effects of CSS, BU, and CY cannot be attributed to an increase in motor activity because they did not induce hyperlocomotion in the open field test. Likewise, imipramine, the positive control, did not alter locomotion, and Bellido et al. (2002) reported a similar finding. Of note, the CMS is reported not to elicit consistent changes in the locomotor response in the open field (Harro et al., 1999). This also suggests that the recovery of sucrose consumption in the stressed animals by CSS and BU in our CMS study was not caused by a locomotor stimulant effect.

It is difficult to compare our results with those of others because CSS and its constituents have not been tested using the FST and CMS model. Nevertheless, it is still possible to learn a great deal from similar studies. Of the two plants that had active effects in our study, we found some useful references related to BU, while there are no scientific data available for CY. Therefore, this discussion focuses on interpreting our data compared with data for similar experiments using herbal preparations that contain BU.

As monoaminergic systems are one of the most important targets in the pathophysiology and treatment of depression and for the action mechanism of the two animal models, the influence of BU on these systems should be considered. A single or repeated dose of Chaihu-Guizhi-Ganjiang-Tang, an herbal compound containing BU, was reported to increase the levels of homovanillic acid (a dopamine metabolite), 5-hydroxyindole acetic acid (a serotonin metabolite), and serotonin in discrete brain regions of mice (Itoh et al., 1996). The aqueous extract of BU is reported to bind dopaminergic and serotonergic receptors (Liao et al., 1995). These reports together with other recent studies showing the important roles of dopaminergic neurotransmission in the effects of the FST are in concordance with our results (Gutierrez-Garcia et al., 2003; Li et al., 2003; Nakazawa et al., 2003).

Several Japanese studies suggested that Saiko-ka-ryukotsu-borei-to, which contains BU, has antidepressant properties (Koshikawa et al., 1998; Mizoguchi et al., 2002, 2003). This herbal medicine ameliorated stress-induced immobility and biochemical changes. Mizoguchi et al. (2002, 2003)

postulated that preventing the attenuation of glucocorticoid feedback is one of the major mechanisms of its antidepressant activity. The oral administration of saikosaponins, active ingredients of BU, increases the plasma adrenocorticotrophic hormone (ACTH) and cortisol levels in rats (Yokoyama et al., 1981). Mizoguchi et al. (2002, 2003) suggested that glucocorticoids secreted by saikosaponins enhanced the feedback mechanism vulnerable to stress. Ayensu et al. (1995) reported high glucocorticoid levels after CMS exposure in rats, while other authors have not found elevated levels of this hormone after prolonged stress situations (Azpiroz et al., 1999). Moreover, Rittenhouse et al. (2002) suggested that the increased behavioral immobility and exaggerated secretion of stress hormones by Wistar Kyoto rats during swimming stress was partly explained by the reduced sensitivity of glucocorticoid receptors that supplied negative feedback to the hypothalamic-pituitary-adrenal (HPA) axis. Therefore, we conjecture that CSS and BU show antidepressant effects in the FST and CMS model owing to modification of the disturbed HPA axis.

γ -Aminobutyric acid (GABA) might also be involved in the mechanism of CSS and BU action in our experiment. It has been suggested that BU activates the GABA receptor (Sugiyama et al., 1996), and ethanol, a GABA agonist, had anti-immobility effects in the mouse FST, which were potentiated by GABA agonists (Hirani et al., 2002). Further experiments, including ligand-binding, electrophysiological, and histochemical studies, are necessary to verify the mechanism of action of CSS and its constituents in our study.

Few studies have examined the influence of CY on the central nervous system (CNS); the anti-immobility effect of CY that we demonstrated may be a novel finding. Although we did not find an antidepressant-like effect of PA, PA is reported to influence processes related to the mechanism of antidepressant action (Song et al., 2001; Dong and Xu, 2002; Liu et al., 2002). At least theoretically, several factors could contribute to producing false negative results in our study. While the reliability and validity of the FST and CMS model are generally accepted (Borsini and Meli, 1988; Wilner, 1997), the effects of some factors on animal experiments should be considered: the sensitivity to the experimental setting can differ according to the strain studied, the animal supplier, and differences in the procedural details. A strain difference was demonstrated for the FST using rats. Sprague-Dawley rats were more responsive to antidepressants than the Wistar-Kyoto strain, which have been proposed as an animal model of depressive behavior (Lahmame et al., 1997; Lopez-Rubalcava and Lucki, 2000). Therefore, the use of SD rats is unlikely to result in a serious false negative, which is a frequently discussed point related to strain differences (Cryan et al., 2002). Acute treatment is widely used in the FST because many antidepressants used for acute treatment are active in the test and because one of the most important merits of the FST is that it is a simple screening tool. However, some antidepressants show superior effects with long-term administration. The 7-day dose regimen used in our study might not be a sufficient duration of treatment, although the antidepressant fluoxetine was effective in the FST after a 7-day treatment, but not after acute treatment (Vazquez-Palacios et al., 2004). While we adopted Porsolt's classic method, other researchers have used modified methods to enhance sensitivity (Cryan et al., 2002). Further investigation using a 14- or 21-day regimen and a modified paradigm might enhance the sensitivity of this screening.

Different drugs should be compared using the same route of administration. However, CSS is normally administered orally, while intraperitoneal injection is preferred to oral feeding for the FST owing to the ease of administration. Imipramine injection was chosen as a positive control and as a reference to other FST studies. Further investigation of the pharmacokinetics of CSS is required in order to resolve this problem.

Our study has intrinsic limitations common to the pharmacological screening of herbal compounds. First, bioactive substances in a plant may interact, thus concealing or exaggerating each other's effects in the test. Other confounding factors, including the place of origin, duration of cultivation, and method of processing the plants, might influence the results of experiments with herbal compounds. To resolve these problems, it is necessary to identify the active materials purified from each plant.

In conclusion, our results suggest that Chaihu-Shugan-San, an oriental herbal medicine, exerts antidepressant-like effects comparable to those of imipramine in experimental animal models. *Radix Bupleuri* and possibly *Rhizoma Cyperi* appear to be the plants that contain the active constituents of Chaihu-Shugan-San. Further research elucidating the action mechanism of these effects using purified ingredients will yield insight into the usefulness of this herbal remedy in the treatment of depression.

Acknowledgments

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